

The Journey towards Responsible AI

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In this module, we are introduced to the origins of responsible AI and learn about organizational approaches to responsible AI. The module walks us through five rules to enable an organization to have a smoother and more successful journey towards responsible AI. While these rules generally focus on the product level, they have implications for organizational levels as well and offer key insights into building effective and empowered teams in this field. The module concludes with key learnings from Axente's experience in the revolution that is responsible AI both at PwC and at client organizations.

Lecture Transcript

0:02 Hello, everyone. My name is Marie Axente. I work for PwC in the UK. And in the last five years, I was very fortunate to be part of a new revolution, a revolution that's called responsible AI. And today in our session, I'm going to share with you a bit of my experience of going in this exciting journey. And what are the key rules in organization every organization should consider following if they want to embark on this journey and have a journey that is unforgettable and fulfilling. Well, let's start our journey understanding why are we talking about responsible AI and where the need for responsible AI is coming from. In the last few years, especially before and during the pandemic, our news outlets, our screens, our social media have been filled with headlines like that. Out of the blue, we suddenly have seen that technology or data have been used, abused or misused in various formats. And the dangerous that the dangers of AI that on one hand have been so heavily portrayed in Hollywood narratives, but also have been picked up by famous researchers or business people are not in fact, dangers that are somewhere far in the future, but there are real and present dangers. And those headlines and many others that we have seen in the media come to confirm that the way technology has been developed so far, has a certain level or a lack of responsibility, where data or algorithms or any other components have been used or have been misused. And the result was a damaging one. It's either we have created technology that has discriminated people of color women, or we have created tools that are now in a position to threaten our fundamental democratic values. We are in a position to understand the real profound and, and big dangers of technology, especially AI. In the same time, we have seen another type of crisis that has unfolded in the last two years in the world, and that was the pandemic. And during the pandemic, we all had to rely heavily on technology so much so that technology have been very much upgraded to be a critical infrastructure asset around the globe with many of us working from home with a lot of data being generated. Many organizations have very quickly understand that the only part toward success and sometimes even survival is to continue to digitize. And in this context, many of them have actually started to picking up what they have learned in the last few years alone about the potential of AI and and the fact that AI has gone from being an emerging and exciting technology that had little proof or or demonstratable impact at scale, to be a technology that is more and more reliable and can deliver results. At the end of 2020, we conducted the survey across multiple geographies, to understand what is the appetite of our clients and their perception of our responsible AI.

AI ETHICS

4:19 It was really exciting to see that responsible AI have moved from being an exciting topic to be debated and discussed on various stages, various events or in the media from being an important business priority. So much so that a lot of our executives, the executives of interviews have confessed, they are looking into having a strategy of AI that's very much aligned with their business strategy. They're looking to further invest in technologies like AI. But in the same time, being acutely aware, there is much more to this technology that it was delivered first being that risk associated and they need to be treated accordingly. Specific capabilities need to be built within the core companies, so that when prompted, with the implication of this technology, they're being addressed on a sustainable and constructive manner over time.

5:32 All this the headlines that we have seen in the media and obviously with the pandemic, where many of us have have used this time to reflect upon upon where the world is going, where technology is being, how technology being embedded in the day to day lives, have prompted a raising awareness of for for not just the benefit of technology, but also the dangers. And we have seen more citizens willing to be part of the conversation, questioning and pushing back when technologies have impacted their own life, as we've seen in the UK, with the A Level results, when an algorithm has awarded students grades based on historical performance of previous colleagues. We've seen a wave of protests in the streets of London against this misuse and underuse of technology that have prompted even more an urge, an urgency, from companies, governments and civil society to work together towards providing the right guardrails. So the technology not only delivers upon the promises, but also doesn't harm people and society as a whole. On the back of those, those increased awareness and headlines as government and regulators have been very, very busy. This, this chart shows a snapshot of the many initiatives that have been conducted across the world by various governments and their associated organizations with the view of, of shaping the narrative around how we develop and use technology, how we use data, but also being able to collectively agree what sort of a guardrail, what sort of a governance is required, what sort of laws and regulation exists or needs to be created in order to deal with those implication of technology and how to do it in a way that doesn't doesn't hamper innovation. Similarly, at organizational levels in industries, we have seen a rise in interest of companies wishing to, to approach AI in a different way. Some of the companies have been in the middle of those incidents that have been reported in the press. And that has almost prompted the peers in the industry to a rethinking of even further, how AI is contributing to the success of their organization or has the potential to hamper its effort. So much so that this perfect storm has prompted experts, AI experts, in the field to conclude that we now have reached a turning point where AI has not only demonstrated its benefits, but also its risks. And we cannot afford to ignore those risks going forward. Hence, if we are really in that unique moment and we are aware of the risks posed by AI, how can we proceed? How can we go forward in a way that we acknowledge that it is not just about benefits, it's also about risks that need to be mitigated, proactively, and what sort of body of knowledge what sort of disciplines we have around disposal to allow us to proceed forward in this journey.

9:18 And very quickly, we came up to the understanding that we actually have a body of knowledge that's now emerging more and more that allows us not only to understand and create a vision of a good life with AI, but also understanding how can we translate this vision in reality, and that's where we resorted, there's a community on the new field called, or the field of AI ethics, that have emerged in the academia from the the brilliant research that has been conducted in the field of technology and computer data in AI ethics across the time. While in parallel in industry, we've seen an emerging of approaches and strategies to be able to pick up on those visions and action upon those issues in a proactive and agile manner. And that's how the two disciplines have been created. And it's

AI ETHICS

really important for us to have this distinction in understanding what is AI ethics and what's responsible AI because at times within the community and outside the AI community, those terms tend to be conflated. While AI ethics is a term preferred by the research, the academia, the civil society, the activists in the industry, we prefer very much more to actually approach those issues via the angle of responsible AI. And the connection is while ethics allow us to continuously iterate on a vision of a good life with AI, responsible AI allows us to pick up on this vision and be able to act upon it, being able to translate those abstract principles into instructions, into design requirements. into governance requirements that will guide those who actually are building or using this technology. And being able to cascade this body of knowledge into something that's more practical and relevant in the day to day life. Why I'm stressing this differentiator is that from our experience, we have seen, working with our internal AI teams and with our clients, the necessity to be clear on the terms we're using, to be clear on the concepts that they are being utilized in order to define a specific context and with that set of actions and decisions. Furthermore, I would say that there's a need to clarify what's the relationship between AI ethics and human rights, also, ethics and risk management, and very importantly, the role of ethics in global compliance. And while the purpose of this session is not so much about discussing definition and framing those concepts, I will encourage you to check the AI C-suite toolkit developed by World Economic Forum with a team of global experts in the field. They have tried not only to contextualize the AI ethics in the context of businesses and how it's linked back to the wider concept of business ethics, bioethics, but being able to create a shared understanding of the concepts and how companies need to be working with those concepts in a day to day perspective.

13:16 Back to our journey. So if you are like us and our clients, and are committed to building and using technology in a way that benefits society, and helps solve important problems, responsible AI became big becomes paramount in importance, becomes as a building block required for this vision to be realized. And from our experience, building a responsible AI toolkit and shaping the narrative both internally and externally, working with partners across the academia, the civil society, and in government in many different parts of the world, we came up with, I think, few rules that allow any organization regardless of the size, to be able to have a more enjoyable and fruitful journey towards responsible AI. And let's pick up one at a time.

14:21 One of the first rules and keep in mind those rules, although they are numbered, I wouldn't say that number one has a higher priority than number five, I would stress that all those rules are important and some of them more important than others, depending on where your organization is and what's the speed of travel toward responsible AI. But I will start with one. You might think that being precise about what the AI technology being used, and what's the context of use, it's common sense. Well, well, if we'd be common sense, perhaps we would have solved many of the, prevent many of the, misuses we have seen. So I would say with confidence, start by being clear about the problem you're trying to solve, and what sort of AI technologies you're using to solve the problem. And be very precise, sometimes we get hung out into the term artificial intelligence, and the fact that no one has a clear definition. And if we are to look at the challenges the European Commission has, in defining the term itself, clearly, we will require a bit more thinking. And maybe that's out of scope for you. So to be very precise, and to be clear, you know, be clear, be precise on how you define a problem, how the problem is linked with your strategic objectives and then from then, decide what's your technology stack you're going to use in solving their problems. Either its label is AI technology, or it's called different. It's less of important. What is important is to be precise, where you're trying to solve it, and what are the tools available for you to be able to solve the problem.

AI ETHICS

16:21 The second rule is to acknowledge that building and using AI, it's a journey, a journey that starts much earlier than defining a problem. And in fact, it doesn't really end with deployment. With the stochastic nature of AI, there is the need for continuous monitoring, almost real time monitoring, of this application performance. While on the other hand, looking at the rich activity of regulators and governments around the world, there will be many, many rules coming from industry regulators, or country wide or a national organization, they will all have to be harmonized at organizational level and apply sometimes, all in the same time. Before the start of developing an application, it's worth acknowledging that the context of development as is the context of use will have a significant impact on how the application will be developed and used, because organizations are using AI in so many different ways. We've seen an absolute explosion of use cases across the value chains of organization, across industry. Therefore, being able to acknowledge that the problems AI is said it's poised to solve need to be linked with strategic objectives. And in solving those problems, how the culture is crafted, what sort of a prioritization of investment, for example, with policies and processes exists, will fundamentally alter how this technology, how AI technologies, are being used, and then are being deployed and designed in such an organization. So in your journey, make sure that you understand what comes before developing an application or comes after. And acknowledge the impact, acknowledge that there's each organization has to have a solid organizational foundation in order to be able to deal with obligations under the AI umbrella.

18:58 I'm a bit chicken I will say, I know what I said earlier, the fact that I'm not going to prioritize the rules for a responsible AI journey, but I think I will pick up rule number three as my favorite. Because the secret is in having a team that is empowered, that is knowledgeable, that has a high level of inclusivity and collaboration to be able to go and investigate issues, to be able to apply the rules but also challenge those rules and think beyond those rules in order to achieve results that are acceptable and with a set of standards, and could be described with labels like fair, transparent, just, equitable, beneficial. And having a team that functions like this requires an understanding that having just technical skills in your team, it's not enough. It's very much like assuming you will play a Beethoven symphony only with pianos. And we all know that, in order to have that fullness of a symphony, like number nine or number three which is my favorite, you require so many different instruments that need to be coordinated so that they bring to life that richness of a sound, the richness of the music that Beethoven has created. I would say that similar to having an A team, you have to be able to rely on your data engineers and data scientists, but you also have to be able to make ethicists as main performer in your orchestra. You have to be able to have legal and social and human scientists that will come and advise you and contribute to how you will elaborate the problem, and then find a solution that really acknowledges for various flavors and impact that AI within its own agency can deliver in a certain context. And in the same sphere of building and maintaining an A team, an orchestra, they will play on tune. It's important to rethink the boundaries of our responsibility. It is not enough to push down on data scientists to solve or address issues of bias in AI. It's very important to acknowledge that the responsibility of how AI enters and also enters an AI lifecycle, but also how it's mitigated is not just down to one group of individuals, it's a shared responsibility that needs to be acknowledged. And with that, a shared solution. So looking at how each member of the team plays their role, and what sort of responsibility you have, takes away the sole pressure from a certain number, or a certain focus on certain job roles, to solve the issues related to implications of AI and allows you to perform a more distributive approach. Similarly, organizational level, it's the moment for companies to acknowledge the role they play in society, the fact that with their activity can have an impact and it's been shown so strongly in the last few years, the impact we all have on our environment and the planet, but also on societies in general. So it's looking at the boundaries of responsibility. It's

AI ETHICS

about reassessing those on a personal level, but also organizational level. Allowing companies to be more honest about the roles and the impact they have in society and for the planet as a whole.

23:39 Moving forward. It's the fourth rule that is looking very much into the fact that any journey might have bumps on the road. Any journey can actually have a certain level of danger. Having said that, for any journey, if we know that risk upfront, we are able to prepare for it. And that's why recognizing the risk associated with AI becomes paramount. And while technology risks in the past have been treated as a sideline set of activities and it has been delegated to compliance or third parties, it becomes really important for AI risk to be considered in a different manner. Being able to understand the risk of bias and opaqueness, it's not something that should be addressed at the end of the life cycle or in the middle, but it's a continuous process. And the mitigation circumstances, mitigation action, is a set of intervention to be designed by both data scientists and compliance professionals will allow organizations to be much more agile how did they respond those risks, but also really important how do you respond to regulatory demands. And in fact, the famous AI Act, the European Commission is working at the moment, is a risk based approach. And whether your business is or not in Europe, more likely that this act will have a profound impact on the thinking and approach of other regulators around the world so expect that more regulators will be following a similar approach. Hence, developing this agility to respond to risk internally will allow you not only to be better in complying with regulation, but to a certain extent be ahead of your competitors.

26:00 And we're coming to the last rule for our responsible AI journey. And yes, this is also one of my favorites. I shouldn't have any favorites, but guess what it looks like I have. It's the land of the promise. So ultimately, we want to make sure that the vision of a good life with AI is realized. And in that vision of a good life, we are able to deal with any unintended negative consequences. And in the same time, we want this technology to reflect our set of values. And in order to do that, we need to be able to translate those values that are quite abstract in nature into principles and then design and governance requirements across the lifecycle. So that we ensure that at the end, when we deploy those AI applications, not only that there is a proof of ability, or the possibility to trace back how a specific values is brought to life in a feature of of the application, but also to be able to prove back either to regulators or whomever needs to be proven to that the technology is aligned with our values. And everything we do, I think, in our organization, in terms of living up to our mission and vision and values, we have to apply the same thoroughness with the technology. And the fact that AI has brought a new set of challenges, it's also a big opportunity for us to be revising how well we embed, how well we live up to our organizational values, and push ourselves and challenge ourselves to do a better job when it comes to technology. Because ultimately, technology we are building it's a fine reflection of a state of our organization. And a good organization will build and use good technology. Similarly, an organization that has multiple issues will end up creating a technology that will reflect those issues. So it's really important for us to find the best ways and the best modality to translate the values and being able to demonstrate and bring evidence on how those values are being enacted. If in your organization, your highest value is integrity and honesty, then it's really important to build transparency and explainability in your application, being able to understand what is the fairness construct that needs to be used in very specific use case, what's the level of inclusivity that you provide, inclusivity and equal access to opportunities and resources, via your products and services. And with that being able to play that important role of changing or addressing the societal problem in inequality.

AI ETHICS

29:59 Well, you might say that those are five, either too simple rules, or maybe are too complex, but the reality is, is that by picking up, maybe not just all of them, maybe a few of them will make a big difference. A big difference that will accelerate the journey and will allow for the journey to be smoother. I'm not going to say that the responsible AI journeys are gonna be smooth sailing but will allow it to be smoother and also much more sustainable going forward. And all those rules are applicable, mainly at a product level. It's fair to say that they have an organizational angle in the back. The fact that those rules applied at product level consistently and across the multiple or the family of AI applications you your organization is building will ultimately trigger or highlight the need for organizational capabilities via processes and people, roles and policies that need to exist in a way that those issues will become business as usual, will become part of how we build and use AI going forward. From, from our experience, applying those rules for ourselves in PwC, but also with our clients, we acknowledge that their organizational level, you need to have a collection of capabilities that will make the biggest difference in your responsible AI journey. And regardless, you start by strengthening your risk management or developing an ethically aligned design process, ultimately, you will have to bring experts in the room, ethicists, you will have to be able to use boards and committees to allow you to elicit some of the ethical dilemmas and issues. But in the end, what will make the big difference is the ability to build a culture of ethics where the personal behavior is aligned with those rules. And the whole organization, especially those involved in the process of AI, are very much a part of the same orchestra, an orchestra that is singing the same tune and each of the members of the orchestra are playing their part in delivering responsible technology, which, in exchange being offered in society to clients, or used to solve societal issues will make a profound impact on our societies.

33:02 If I were to think of one major enabler that is applicable in every single context, across the board, across geographies, in relation to how we approach technology, not just AI, I would say that we need a new vision for computer science. For too long, enabled by Silicon Valley's philosophy: move fast and break things, we have built technology irrespective of its implication and impact. It's time now to pause and really consider what those implications are, and move from the philosophy of can we build it because we have all the components in place, and there's little of no constraints into an approach that really considers the consequences and implication of buildings such technology and proceeding with design in a way that those implication are being addressed in a way that we maximize the benefits that AI and its technologies and any other technology stacks are being developed and minimizing the risk. And are all the risk of being known up front? And I'm sure that AI technology has many surprises for us in full, having a solid foundation by having a reliable set of experts that sing the same tune and they are empowered by the right tools and processes and frameworks and enablers will make this disruption or managing on unintended consequences, much smoother than otherwise.

35:26 I really hope that sharing my experience in being part of responsible AI journey for so long will support your journey as well. And you have found my rules or you will find my rules helpful in the future. I would like to leave you with a thought that probably it becomes core to how I approach all the conversations I have in relation to responsible AI. Reflecting upon the potential of AI to disrupt the life as we know it, a profound change is a horizon and we know that humans are not necessary fans of change. And therefore, it's very possible that we will encounter more resistance to change. But in this resistance to change, we need to understand where it is coming from, and especially when we will be met with barriers that seem not easy to break, it's to refuse to accept that that's not possible. There's always a way for us to go forward. There's always a way and an alternative. As long as we find the openness to sit and have a dialogue, a dialogue across different parts of society, and being able to find

AI ETHICS

solutions that will not only empower more groups but also benefit more as well. Thank you very much for having me. I hope you enjoy the session today and see you later. Thank you.